



DOCUMENT ID

D1907-2026-001

CODENAME

JADEPUFFER

DOCUMENT CONTROL	
Classification	OSINT
Threat Level	HIGH
Status	OBSERVED
Type	Autonomous AI Agent
Released	2026-07-01

SUMMARY

Autonomous AI agents may soon replace humans in many offensive cyber operations. JADEPUFFER offers a glimpse of that future.

DOSSIER 1907

THREAT INTELLIGENCE SERIES

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01 EXECUTIVE SUMMARY

JADEPUFFER is one of the first publicly documented demonstrations of an autonomous AI agent executing a ransomware attack chain. During testing, it gained access, harvested credentials, moved laterally, and encrypted a target database without requiring step-by-step human direction.

When an authentication attempt failed, the agent didn't wait for instructions. It identified the error, generated a new approach, and successfully resumed the operation in just 31 seconds.

None of the individual techniques were new. The significance lies in who executed them. JADEPUFFER suggests that autonomous AI agents are beginning to transition from tools that assist cyber operators to systems capable of carrying out complex offensive tasks on their own.

WARNING

JADEPUFFER didn't assist the attacker—it became the operator.

02 KEY FINDINGS

- Planned and executed a multi-stage intrusion.
- Adapted to failed authentication attempts.
- Selected follow-up actions based on the environment.
- Completed the attack chain with minimal operator guidance.

03 TECHNICAL ANALYSIS

JADEPUFFER initiated the intrusion by exploiting CVE-2025-3248, an unauthenticated remote code execution vulnerability in Langflow. Once execution was obtained, the agent immediately transitioned into reconnaissance, enumerating the host, identifying running services, and searching for cloud credentials, API keys, and database connection strings exposed within the environment.

Using the recovered credentials, the agent dumped the Langflow PostgreSQL database to identify additional targets before pivoting into the internal network. It enumerated reachable services, probed MinIO object storage using default credentials, and targeted an exposed Alibaba Nacos instance through CVE-2021-29441 to obtain administrative access. The defining characteristic of the operation was not the exploitation itself, but the decision loop driving it. During the Nacos compromise, the agent generated an invalid bcrypt hash while creating an administrative account. Rather than repeatedly executing the same payload, it analyzed the authentication failure, removed the malformed account, regenerated a valid hash using a different method, recreated the account, and successfully authenticated 31 seconds later. This behavior demonstrated autonomous error recovery rather than scripted retry logic. With administrative access established, JADEPUFFER targeted the production MySQL environment. It encrypted more than 1,300 Nacos configuration items using MySQL's native `AES_ENCRYPT` function, deleted the original configuration and history tables, and created a `README_RANSOM` table containing ransom instructions. Researchers also observed that the encryption key was never retained or exfiltrated, rendering the encrypted data unrecoverable even if a ransom were paid.

04 ASSESSMENT

DOSSIER 1907 assesses that JADEPUFFER marks the beginning of a transition from human-operated intrusions to autonomous cyber operations. While current implementations remain experimental, the pace of AI development suggests this transition may occur far sooner than many organizations expect.

The future cyber operator may not be a person behind a keyboard, but an autonomous agent capable of planning, adapting, and pursuing objectives with minimal human oversight.

05 REFERENCES

1. Clark, Michael. JADEPUFFER: Agentic ransomware for automated database extortion. Sysdig Threat Research, 1 July 2026.
2. MITRE ATT&CK® Enterprise Framework.